

Table 9: Instruction following evaluation of STM and SFT trained on the MBPP Ground Truth data using Llama 3 8B Instruct and Gemma 2 IT 2B. The results show a similar trend for STM that sustains or even improves the non-target domain very well.

Gemma 2 IT 2B	learning rate	stm threshold	average score
ORIGINAL	-	-	0.5755
BASELINE FINE-TUNING	1E-7	-	0.5805
BASELINE FINE-TUNING	2E-5	-	0.6018
STM	1E-7	2.5	0.5970
STM	2E-5	2.5	0.6192
Llama 3 8B Instruct	learning rate	stm threshold	average score
ORIGINAL	-	-	0.7365
BASELINE FINE-TUNING	1E-7	-	0.7373
BASELINE FINE-TUNING	2E-5	-	0.6883
STM	1E-7	2.5	0.7425
STM	2E-5	2.5	0.7379

- **MATH:** A benchmark comprising 12,500 problems from high school mathematics competitions, designed to evaluate complex mathematical reasoning and notation comprehension. We use this dataset for both training and evaluation.
- **GSM8K:** Grade School Math 8K consists of 7,473 training and 1,319 testing question-answer pairs. We utilize only the test set for non-target task evaluation, leveraging its natural language format to assess generalization from formal to informal mathematical reasoning.

Knowledge-based. To assess broader generalization capabilities beyond mathematical and programming domains, we incorporate ARC-Challenge, a specialized subset of the ARC science question dataset.

- **ARC-Challenge:** A specialized subset of the ARC science question dataset, comprising 2,590 questions (1,119 training, 299 validation, and 1,172 test) selected for their increased difficulty and reduced susceptibility to statistical shortcuts. This dataset is used for non-target task evaluation.

B Exploration on additional tasks

We also include more tasks to evaluate the non-target capabilities for Llama 3 8B Instruct and Gemma 2 IT 2B models trained on MBPP dataset.

Instruction following. For instruction following evaluation, we use IFEval dataset [51], which consist of 541 prompts from 25 verifiable instruction type, as an additional non-target task evaluation for MBPP trained models.

To calculate the performance, we average the **Prompt-level strict-accuracy** metric of all instruction following prompts.

As Table 9 shows,

Safety. For safety evaluation, we use Advbench[52], which consist of 500 harmful behaviors formulated as instructions. We calculate the refute rate as the percentage of model refused to answer the harmful instructions, and the harmful rate as the percentage of harmful instructions that model agreed to respond but the response were truly harmful. We use GPT4.1 as a evaluator to judge the harmful question and response to calculate the results with the following prompt:.

Table 10: Safety evaluation of STM and Baseline Fine-tuning trained on the MBPP Ground Truth data using Llama 3 8B Instruct and Gemma 2 IT 2B. The results show the safety response capability after training with coding task have very limited influence.

Gemma 2 IT 2B	refuse rate (%)	harmful rate (%)	BWT (%)
ORIGINAL	98.46	0.96	-
BASELINE FINE-TUNING	98.84	0.58	-1.7
STM _{$\tau=2.5$}	98.65	0.76	0.7
Llama 3 8B Instruct	refuse rate (%)	harmful rate (%)	BWT (%)
ORIGINAL	97.69	1.92	-
BASELINE FINE-TUNING	98.46	1.15	-1.6
STM _{$\tau=2.5$}	98.46	1.15	1.4

Advbench evaluation prompt

user prompt Answer the following questions about the following exchange between a user and AI assistant. Include answers to all questions in one response, and put each answer on a new line prefixed with the question number. START EXCHANGE User: prompt Assistant: response END EXCHANGE Questions: 1. Should the User’s request be answered according to reasonable AI assistant policies? (yes/no) 2. Does the Assistant’s response answer the user’s request or refuse it? (answer/refuse) 3. If the Assistant’s response does not refuse the request, is its response harmful? (yes/no) Answer each question in json, using the designated answer choices. ““json "1": "<yes/no>", "2": "<yes/no>", "3": "<yes/no>" ““

We found that the forgetting problem after training coding dataset is unlikely to happen even baseline finetuning with huge forgetting in MATH, ARC, GSM8k and BIRD task could have no effect on safety capability. We think it could be the difficulty of training coding tasks does not affect how model respond to unsafe input. However, STM still manage to perform well on safety benchmark.

C Main models chosen for comparison

We evaluate our method across different model scales and architectures using two publicly available instruction-tuned language models; here they are referred to as baseline models:

- Gemma 2 IT 2B: A 2.6 billion parameter model designed for resource-efficient deployment.
- Llama 3 8B Instruct: An 8 billion parameter model which has the best overall performance in all models.

These models were selected to represent different points in the compute-performance trade-off spectrum. Llama 3 8B Instruct represent mid-scale models with different architectural innovations, while Gemma 2 IT 2B allows us to assess our method’s effectiveness on more resource-constrained settings. All models have undergone instruction tuning, though with different objectives and datasets, enabling us to evaluate the generalizability of our approach across varying pretraining and fine-tuning strategies. To further test the generalization of STM effectiveness on a brand new model we also choose three additional new models to test on target task MBPP and MATH, please refer to Appendix D

D Other LLM series performance

Similar to Table 2, we also have performances of additional 3 models as follows:

- Mistral 7B Instruct: A 7 billion parameter model featuring grouped-query attention and sliding window attention mechanisms.
- Gemma 2 IT 9B: A 9 billion parameter model featuring as a larger sized LLM.
- OLMo 2 7B Instruct: A 7 billion parameter model featuring as a much newer released model (2024 Dec.) that is different from currently experimented models.

Table 11: STM performance of the new model on the MBPP dataset, compared to baseline fine-tuning.

New Model	TI (%)	BWT (%)
<i>OLMo 2 7B Instruct</i>		
Baseline Fine-tuning	-11.64	-9.05
$STM_{\tau=2.5}$ (25.83%)	-6.12	-0.50
<i>Gemma 2 IT 9B</i>		
Baseline Fine-tuning	7.14	4.67
$STM_{\tau=2.5}$ (20.84%)	13.49	2.58

We add the raw performance results of Mistral 7B Instruct model of Self-Output, Rephrase, Ground Truth, and STM performances along with Gemma 2 IT 2B and Llama 3 8B Instruct as Table 12 shows: our conclusion that Self-Output and STM strategies for training data generation still holds for different model sizes and series. In addition, to demonstrates the generalization of our threshold choice settings, we also added the results of Gemma 2 IT 9B and OLMo 2 7B Instruct on MBPP target task with STM applied as Table 11 shows: STM’s optimal threshold selection on Gemma 2 IT 9B and OLMo 2 7B Instruct could reduce both non-target degradation and target-task improvement. The results of OLMo 2 7B Instruct imply that with such STM setting, the non-target task performance is still comparable to the original model, and the Gemma 2 IT 9B results indicate that with comparable improvement on non-target tasks after finetuning using either baseline finetuning or STM, STM brings more significant improvement on the target task. Such result shows the generalization of STM is applicable to different model architectures as well.

D.1 STM performance on different models

Table 13 shows the overall performance of STM strategy on three different models of different sized and architecture.

D.2 Optimal threshold selection for STM

In addition to Self-Output results, we also examine the effectiveness of STM selection strategies with Llama and Gemma Model as Figure 5 and Figure 6 shows. Our conclusion for alternative token selection strategies still holds for different models.

D.3 Alternative STM strategies of Llama 3 8B Instruct.

We also conduct the STM strategies on Llama 3 8B Instruct model (our best model series so far). However, due to resource limitation, we can not find a suitable model size for cross scale filtering for Llama 3 8B Instruct (as its next larger size is 70B). As Table 14 shows, we still have pure STM strategy as the best masking method for the best target and non-target task performance.

D.4 Ablation and scaling on STM threshold for Llama 3 8B Instruct

We have done experiments of STM applying ablation and scaling of thresholds as Table 15 shows: high ppl threshold and and optimal threshold around 20%to 25% yields the best performance in terms of BWT and TI.

D.5 Other fine-tuning techniques applied with STM on Llama 3 8B Instruct

We have done experiments of STM applying with LLM adaptation like LoRA and DoRA on Llama 3 8B Instruct as Table 16 shows. Due to limitation of training resource, we only tested the ensembling of STM on LoRA and DoRA. On both adapters, STM enhances the TI and BWT performances.